

RieszNet and ForestRiesz: Automatic Debiased Machine Learning with Neural Nets and Random Forests

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SUMMARY

- Goal: Estimate **linear functionals** of high-dimensional/nonparametric regressions with **$\sqrt{n}$ -consistency** via automatic debiasing.
- Key idea: Learn the **Riesz representer (RR)** from oracle access to the functional; no analytic form needed.
- Methods: **RieszNet** (multitask neural net) and **ForestRiesz** (GRF-based locally linear RR).
- Properties: **Double robustness**, Neyman orthogonality, semiparametric efficiency.

Title Code: github.com/victor5as/RieszLearning

Motivation and Contributions

WHY

- Plug-in estimators suffer from **regularization bias** and model selection effects.
- Debiasing via **RR** enables $\sqrt{n}$ -consistent, asymptotically normal estimation and valid CIs.

THIS WORK

- Automatic learning of **RR** using black-box access to linear functionals.
- **RieszNet**: multitask NN minimizing combined RR and regression losses with targeted regularization.
- **ForestRiesz**: GRF that learns a locally linear RR, stable in continuous treatments.
- Strong empirical performance on **ATE** and average derivatives vs. state-of-the-art plug-ins.

Abstract Semiparametric efficiency; double robustness

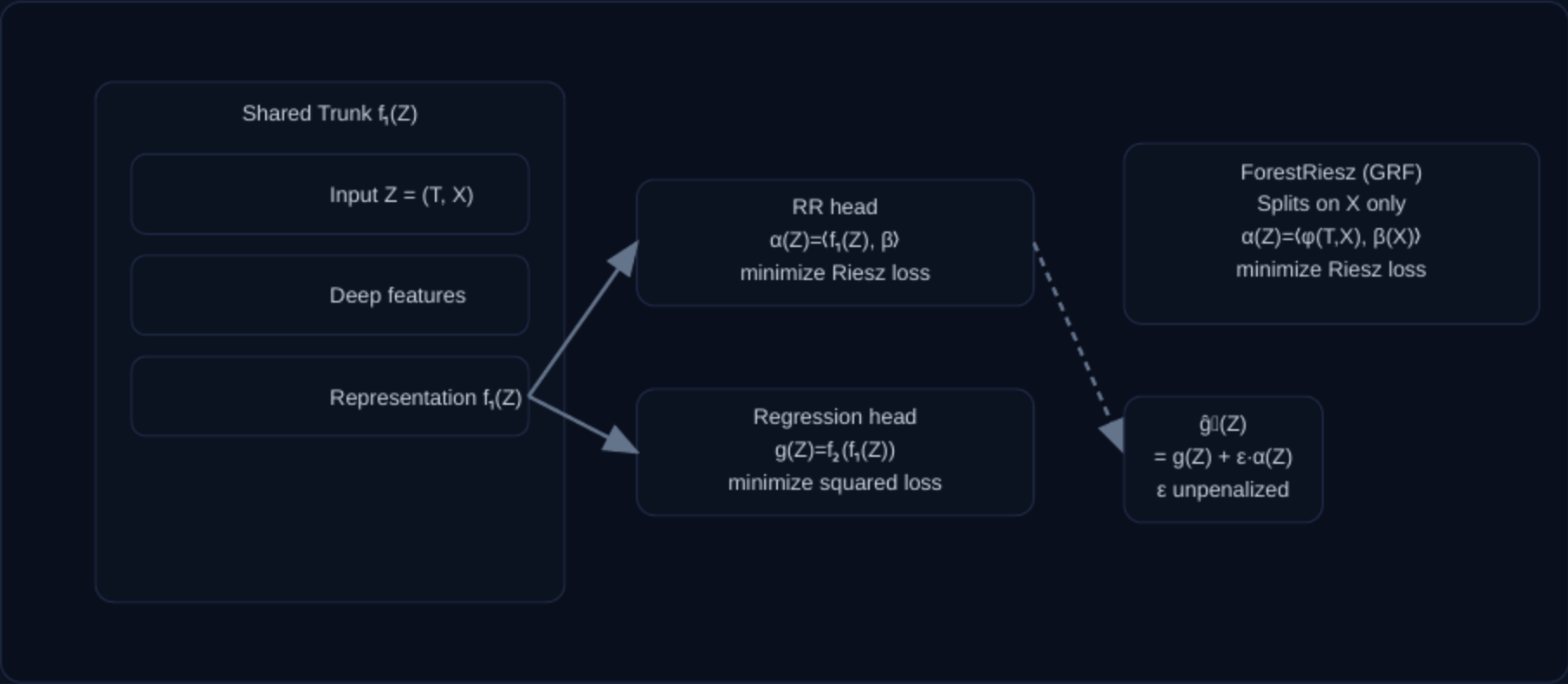
Most Surprising Finding

Standout result: Our fully generic **Auto-DML** approach (RieszNet/ForestRiesz) *matches or outperforms* state-of-the-art **plug-in** methods even when the analytic **RR** is known (e.g., ATE on IHDP).

Why unexpected: Plug-ins exploit known formulas (e.g., propensity-based RR), yet direct **RR learning** via loss minimization is **more stable** and **better calibrated** in finite samples.

Highlight Generic learner beats specialized plug-ins

RieszNet and ForestRiesz: Architecture Overview



Method Start with the big picture

Problem Setup: Average Moments of Regression Functions

SETTING

- $W=(Y,Z)$, $Z=(T,X)$; target $\theta_o = E[m(W; g_o)]$, $g_o(Z)=E[Y|Z]$. Examples: **ATE**, policy value, avg. derivative, incremental effects.

DEBIASED SCORE

- RR $\alpha_o(Z)$: $E[m(W; g)] = E[\alpha_o(Z) g(Z)]$. Score $\psi = m(W; g) + \alpha(Z)(Y-g(Z)) - \theta$.
- Mixed-bias $\Rightarrow$ double robustness; $\sqrt{n}$ inference under product-rate condition; use **cross-fitting**.

Background

Cross-fitting reduces overfitting bias

Examples of Linear Functionals

ATE (BINARY T)

- $Z=(T,X)$; unconfoundedness; RR: **$T/p_o(X) - (1-T)/(1-p_o(X))$** .

POLICY VALUE

- Offline evaluation for $\pi:X \rightarrow \{0,1\}$ via linear moment; doubly-robust learning.

AVERAGE DERIVATIVE

- Continuous T; RR via integration by parts using joint density f_o .

INCREMENTAL EFFECTS

- Infinitesimal $\pi:X \rightarrow [-1,1]$; useful for policy gradients.

Examples

RR exists under mild conditions

Estimator and Asymptotics

ESTIMATOR

- $\theta_n = E_n[m(W; \hat{g})] + E_n[\hat{\alpha}(Z)(Y - \hat{g}(Z))]$; bias shrinks with product of L2-errors of $\hat{g}$ and $\hat{\alpha}$.

INFERENCE

- Under product-rate $> n^{-1/2}$: $\sqrt{n}(\theta_n - \theta_o) \Rightarrow N(0, E[\psi_o(W)^2])$; ψ_o achieves efficient variance.

Theory

Semiparametric efficiency

Learning the Riesz Representer via Loss Minimization

CORE IDEA

- Riesz loss $L(\alpha)=E[\alpha(Z)^2 - 2 m(W; \alpha)]$ minimized by α_o ; empirical minimization over A_n avoids analytic RR.

RATES

- Fast rates via critical radius δ_n for A_n and $m \circ A_n$; mean-squared continuity ensures stability.

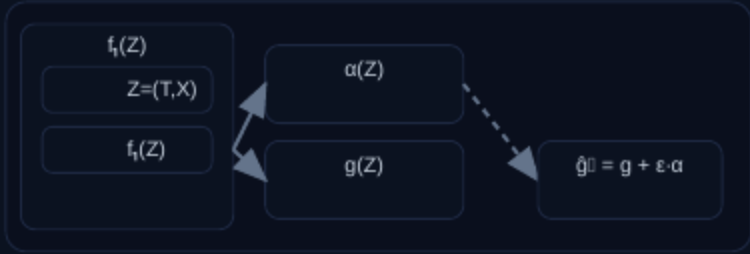
RR Loss

Bias–variance trade-off tailored to RR

RieszNet: Multitask Neural Network

DESIGN

- Shared $f_1(Z) \rightarrow$ RR head $\alpha(Z)=\langle f_1(Z), \beta \rangle$ (Riesz loss) and Regression head $g(Z)=f_2(f_1(Z))$ (squared loss).
- Targeted regularization: $\hat{g}(Z)=g(Z)+\varepsilon \cdot \alpha(Z)$, ε unpenalized; joint loss = RR + regression + targeted + reg.



Key Lemma: Central Role of the Riesz Representer

LEMMA

- It suffices to estimate $g_o(Z)=h_o(\alpha_o(Z))$ with $h_o(A)=E[Y|A]$; prioritize features predictive of $\alpha_o(Z)$.

TARGETED REGULARIZATION EFFECT

- FOC: $E_n[(Y-\hat{g}(Z)) \cdot \alpha(Z)]=0 \Rightarrow$ DR plug-in equivalence; bridges one-step and TMLE.

ForestRiesz: Locally Linear RR via GRF

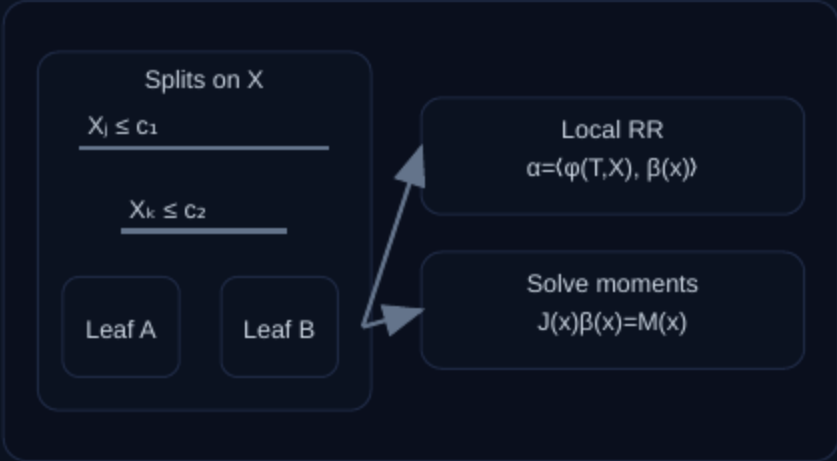
CHALLENGES

- Splitting on T can break mean-squared continuity and ill-pose local problems.

APPROACH

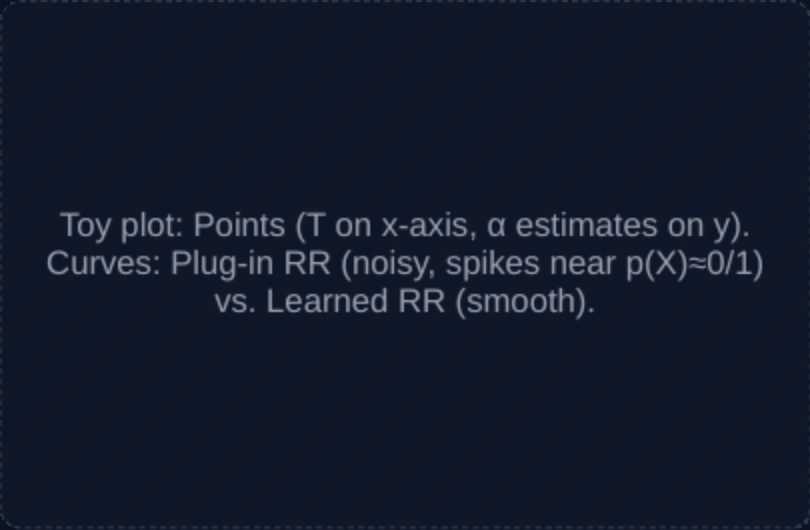
- Split only on X; model RR as $\alpha(Z)=\langle \varphi(T,X), \beta(X) \rangle$ with smooth φ .
- Estimate $\beta(x)$ via GRF solving local moment conditions; split criterion minimizes Riesz loss, penalizes ill-conditioned Jacobians.
- Multitask option: include regression moments.

SCHEMATIC



Qualitative Illustration: RR vs. Plug-in Behavior

TOY 2D VIEW



WHAT TO NOTICE

- Stability**: learned RR smooths extreme weights when propensity near 0/1.
- Bias–variance**: direct RR learning controls variance without requiring explicit $p(X)$ inversion.
- Downstream**: more stable RR $\rightarrow$ tighter, better-calibrated DR CIs.

Experiments: Overview

ATE (IHDP)

- 1000 semi-synthetic datasets (setting A).
- Compare **Direct**, **IPS**, **DR**; RieszNet/ForestRiesz vs. plug-ins.
- Metrics: MAE, **bias**, **RMSE**, coverage.

AVERAGE DERIVATIVE (BHP GASOLINE)

- Semi-synthetic from 3,640 households; various DGPs.
- Assess **DR** with/without TMLE adjustment; **cross-fitting**; multitasking.

Experiments

DR performs best and achieves near-nominal coverage

Results: IHDP ATE

RIESZNET (MAE ± SE)

- Direct: 0.123 ± 0.004
- IPS: 0.122 ± 0.037
- DR: 0.110 ± 0.003**

BENCHMARKS

- Dragonnet (plug-in): 0.146 ± 0.010

FORESTRIESZ (MAE ± SE)

- Direct: 0.197 ± 0.007
- IPS: 0.669 ± 0.004
- DR: 0.126 ± 0.004**

BENCHMARKS

- RF Plug-in: 0.389 ± 0.024
- CausalForest: 0.728 ± 0.028

COVERAGE

- With redraw of T: **DR CIs** cover ≈95–96% (nominal 95%).
- Direct** underestimates variance → poor coverage.

IHDP

Auto-DML outperforms plug-ins

IHDP: Distribution, Bias, RMSE, Coverage

Figure Placeholder: RieszNet and ForestRiesz performance plots over 100 datasets (bias, RMSE, coverage, estimate histograms)

IHDP

DR centered near truth; approx. normal

BHP Gasoline Demand: Setup

DATA GENERATION

- Model conditional T|X via RF: $\mu_{\tau}(X), \sigma_{\tau}^2(X)$; draw $T \sim N(\mu_{\tau}, \sigma_{\tau}^2)$.
- Generate $Y = f(T,X) + \varepsilon$; designs: **simple/complex f**; linear/nonlinear X.
- Targets average marginal effect; truth ≈ −0.6 in simple designs.

BHP

Assess stability under continuous T

Results: BHP Average Derivative (Complex f, linear+nonlinear X)

RIESZNET

- DR: Bias 0.062, RMSE 0.504, Cov 0.877
- Direct: Bias 0.053, RMSE 0.562, Cov 0.056
- IPS: Bias 0.061, RMSE 0.496, Cov 0.916

FORESTRIESZ

- **DR + post-TMLE: Bias −0.082, RMSE 0.327, Cov 0.953**
- DR: Bias −0.131, RMSE 0.377, Cov 0.909
- Direct: Bias 0.094, RMSE 0.304, Cov 0.046
- IPS: Bias −0.161, RMSE 0.443, Cov 0.847
- RF Plug-in: Bias 0.055, RMSE 0.327, Cov 0.912

BHP

DR (+TMLE) yields best balance of bias & coverage

BHP: Distribution, Bias, RMSE, Coverage

Figure Placeholder: RieszNet and ForestRiesz distributions and CI coverage across seeds and DGPs

BHP

Approx. normal estimates; small bias under DR

Ablations: RieszNet (IHDP)

FINDINGS

- **Multitasking** shared trunk crucial: separate NNs increase bias/RMSE and reduce coverage.
- **No end-to-end** (freeze shared layers) → much higher variance and lower coverage.
- **TMLE post-processing** similar coverage to end-to-end targeted reg., but slightly worse bias.

Table Placeholder: Bias/RMSE/Coverage across variants (Baseline, Separate NNs, No end-to-end, TMLE post-proc.)

Ablation

End-to-end multitask learning matters

Ablations: ForestRiesz (BHP)

FINDINGS

- **Simple cross-fitting** markedly improves CI coverage.
- **Multitasking** GRF helpful; double cross-fitting incompatible with multitask sharing.
- **DR + post-TMLE** generally best bias–coverage trade-off.

Table Placeholder: Bias/RMSE/Coverage for (no x-fit/m-task, no x-fit/with m-task, x-fit/no m-task, double x-fit)

Ablation

Cross-fitting supports valid inference

Practical Guidance

- Use **DR estimator** with **cross-fitting** for inference.
- Prefer **multitask** shared representations for g and α .
- For continuous T , enforce smooth **$\varphi(T,X)$** and **restrict splits to X** (ForestRiesz).
- **Targeted regularization** ($\varepsilon\text{-}\alpha$) stabilizes plug-in and aligns with DR score.

Key Objects (Schematic)

DEBIASED SCORE

$$\psi(W; g, \alpha, \theta) = m(W; g) + \alpha(Z) \cdot (Y - g(Z)) - \theta$$

RIESZ LOSS

$$L(\alpha) = E[\alpha(Z)^2 - 2 \cdot m(W; \alpha)] \rightarrow \text{minimize over } \alpha \in A_n$$

Conclusions

- Automatic **RR learning** enables generic debiasing for linear functionals.
- **RieszNet** and **ForestRiesz** achieve strong accuracy and near-nominal coverage.
- **End-to-end** multitask learning and **cross-fitting** are key.
- Approach complements theory of Auto-DML with practical NN/GRF implementations.